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RESEARCH ARTICLE

Designing an Optimal Washout Filter for a Flight Simulator Using Linear Quadratic Regulator (LQR)

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Abstract

This study presents a motion cueing algorithm (MCA) to enhance the fidelity of simulator motion platforms limited by physical constraints. The MCA aims to accurately reproduce vehicle accelerations and angular velocities to provide a more realistic motion experience for human-in-the-loop applications, such as driving and flight simulation. The proposed approach employs a higher-order optimal washout filter based on Linear Quadratic Regulator (LQR) that takes into account the vestibular system mathematical model, simulator constraints, and capabilities. The filter is designed to reduce human perception error between the simulator motion platform and the simulated vehicle, using angular velocity and linear acceleration as inputs. The obtained washout filters are optimized transfer functions that link the simulator motion input to the vehicle motion input, aiming to constrain the human sensation error and the platform motion. The proposed MCA overcomes the limitations of previous classical washout filters by utilizing human perception, exploiting available workspace more efficiently, and respecting physical constraints. The results show that the proposed algorithm provides a more realistic motion experience for human-in-the-loop applications, without requiring expensive high-fidelity proprietary machinery.

Keywords Motion Cueing Algorithm (MCA), Human-in-the-Loop, Optimal Washout Filter, Linear Quadratic Regulator (LQR), Vestibular System

1. Introduction

Motion platforms are used in human-in-the-loop applications such as driving and flight simulation to immerse users into the simulated experience. However, the operational workspace of the platform is limited compared to the vehicle being simulated, making it difficult to replicate many maneuvers within the simulation environment. To address this issue, the paper proposes using a motion cueing algorithm (MCA) to provide realistic motion cues to users while maintaining fidelity and staying within the platform's workspace [1].

The classical washout filter is a commonly used MCA in commercial simulators, but it has limitations, such as requiring trial-and-error to determine filter parameters [2]. To address this issue, an adaptive washout algorithm was proposed, which incorporates the vestibular system to reduce false cues and improve simulation quality [3] and [4]. However, sub-optimal usage of the motion platform workspace remained an issue.

The optimal motion cueing algorithm was then introduced, based on the human perception of the desired vestibular system dynamics. This algorithm takes into account the limitations of the classical and adaptive washout filters and provides a natural progression towards a more human perception-focused simulation [5].

Optimal control is a strategy with the goal to discover a control rule for a particular system that meets a specific optimality criterion. It has evolved into an organic extension of the adaptive algorithm in the context of motion cueing algorithms. Given the fundamental issue of finding the best way to provide motion signals to maximize the similarity between pilot sensation in the simulator and in reality, the optimality criterion here becomes the sensation equivalence, or in other words eliminating the error between the two.

As discussed in literature, the optimal control-based algorithm approaches motion cueing as a tracking problem, with the goal of minimizing the sensation error between the sensed accelerations on the platform and in the real vehicle while staying within the bounds of the motion platform. This requires that the accelerations experienced in the simulator track the accelerations experienced in the real vehicle as closely as possible. A knowledge of the connection between real bodily motion and motion perceived by the brain is necessary for such a strategy. Higher order filters with optimal control strategies, namely the linear quadratic regulator, LQR, are used in this algorithm. This technique limits the sensation inaccuracy between the simulated car and motion platform dynamics by incorporating a mathematical model of the human vestibular system.

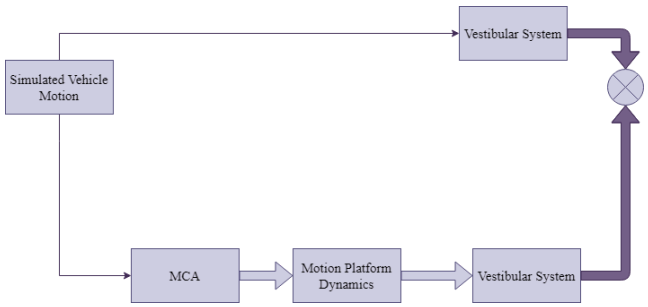

Figure 1 Overview of Sensation Error

2. The LQR controller:

LQR control is an optimal control strategy with quadratic indices. LQR is straightforward and enables closed-loop optimum performance via either output feedback or linear state feedback. LQR calculates the control signals by simulating it as an optimization problem. In other words, it seeks to calculate control signals that minimize some cost function.

LQR generates a steady state minimal error with robust features and regulator properties. Rapidly resolving system problems is another benefit of this technique, in order for the system to retain equilibrium in the face of environmental disruption. This section will briefly touch on the derivation and solution to the LQR problem, which will later be used in the development of a motion cueing algorithm.

The linear quadratic regulator (LQR) with a finite horizon is given by:

$$x^{\cdot} = Ax + Bu\, x \in R^n, u \in R^n, x_0 \quad (1)$$

The cost function is given by:

$$J = \frac{1}{2}\int_0^T \left(x^T Qx + u^T Ru\right)dt + \frac{1}{2}x^T(T)P_1 x(T) \quad (2)$$

where $Q \geq 0, R > 0, P_1 \geq 0$ are symmetric, positive (semi-) definite matrices.

Solving using the maximum principle:

$$H = x^T Qx + u^T Ru + \lambda^T(Ax + Bu) \quad (3)$$

$$x^{\cdot} = \left(\frac{\partial H}{\partial \lambda}\right)^T = Ax + Bu\, x(0) = x_0 \quad (4)$$

$$-\lambda^{\cdot} = \left(\frac{\partial H}{\partial x}\right)^T = Qx + A^T\lambda\, \lambda(T) = \quad (5)$$

$$0 = \frac{\partial H}{\partial u} = Ru + \lambda^T B \Rightarrow u = -R^{-1} \quad (6)$$

For simplification, guess the solution of H in the form of $\lambda(t) = P(t)x(t)$ then:

$$\lambda^{\cdot} = P^{\cdot}x + Px^{\cdot} = P^{\cdot}x + P(Ax - BR \quad (7)$$

$$-P^{\cdot}x - PAx + PBR^{-1}BPx = Qx + \quad (8)$$

The equation is satisfied by finding $P(t)$ which is the solution of the algebraic Riccati equation given by:

$$-P^{\cdot} = PA + A^T P - PBR^{-1}B^T P + Q \quad (9)$$

3. The Optimal Washout Filter:

This section documents the development of an optimal washout filter algorithm based on the work of [6], [7], and [5]. The goal of the proposed optimal washout algorithm is to minimize the cost function by limiting the sensation error between the simulator and platform motion. The problem is to determine a washout matrix of linear transfer functions W(s) that relates the simulator motion input to the vehicle motion input.

Figure (2) shows the algorithmic plan for optimal washout.

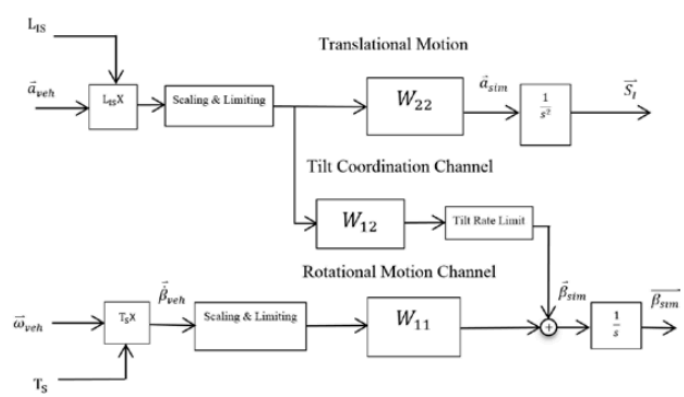

Figure 2 Optimal Washout Filter

the transfer function matrix W(s) is found so that the estimated error between the detected signals of the vehicle and the simulator is as little as possible. The input values are represented by $\underline{u}_a$ in the aircraft and by $\underline{u}_s$ in the simulator (notation a and s will always refer to aircraft and simulator respectively). Their relationship, which also includes the transfer function, is as follows:

$$\underline{u}_s = \underline{W}(s)\,\underline{u}_a, \quad \underline{W}(s) = \left[W_{11}\, W_{12}\, W_2\right. \quad (10)$$

The next step is to choose the most suitable control inputs. As in this study, [6] employed angular displacement, longitudinal and lateral linear acceleration, and both as control input. Input vector therefore has the following shape:

$$u = [\beta \; a] \qquad (11)$$

The vehicle dynamics, and simulator dynamics are all modeled as linear differential equations given by:

$$\dot{x}_a = A_v x_a + B_v u_a \quad y_a = C_v x_a + D_v u_a \; ; \qquad (12)$$

Since the motion of the simulated vehicle cannot be limited to one specific scenario, the input will be modeled as filtered white noise with the following state space representation.

$$\dot{x}_n = A_n x_n + B_n u_n \quad u_a = x_n \quad A_n = \left[-\gamma_1 \; 0 \right. \qquad (13)$$

where the first-order filter break frequencies for each of the degrees of freedom are given by γ_1 and γ_2. The extra internal variables, which are listed in Table (3), can be stated as follows:

$$\dot{x}_d = A_d x_d + B_d u_S \qquad (14)$$

Next the vestibular system, which consists of the otolith and the semicircular canal, as mentioned previously, is represented in state space as:

$$\dot{x}_v = A_v x_v + B_v u \quad y_v = C_v x_v + D_v u \qquad (15)$$

Where u and y_v, in this case, is true for both the aircraft and the simulator.

Finally, the pilot and vestibular sensation errors are given by:

$$e = y_S - y_a \qquad (16)$$

$$x_e = x_S - x_a \qquad (17)$$

$$\dot{x}_e = A_v x_e + B_v u_S - B_v u_a \quad e = C_v x_e + D \qquad (18)$$

The state space representations are all collected and integrated into a single state space model as:

$$x = \left[x_e^T \; x_d^T \; x_n^T \right]^T, \; y = \left[e^T \; x_d^T \right]^T \quad \dot{x} = Ax + B \qquad (19)$$

Where:

$$A = \left[A_v \; 0_0 \quad -B_v \; 0 \; A_d \; 0 \; 0 \; 0^2 \; A_n \right], \; B = \left[B \right. \qquad (20)$$

The cost function, for the LQR controller in order to minimize the sensation error is in the form:

$$J = E\left\{ \int_{t_0}^{t_1} \left[e^T Q e + \rho \left(u_S^T R u_S + x_d^T R_d x_d \right. \right. \right. \qquad (21)$$

where Q, R_d and R are symmetric positive matrices and ρ is a positive scalar value.

the cost function can be transformed to:

$$J = E\left\{ x^T R' x + \rho u_S'^T R_2 u' \right\} \quad \dot{x} = A' x + \qquad (22)$$

Where:

$$A' = A - B_{R_2^{-1}}^{R_{12}^T}, \; u' = u_S + R_2^{-1} R_{12}^T x, \; R_1' = R \qquad (23)$$

The cost function is at a minimum when:

$$u' = -R_2^{-1} B^T P x \qquad (24)$$

Where the matrix P is the solution to the algebraic Riccati equation as explained briefly above:

$$0 = R_1' - P B_2^{-1} B^T P + A^T P + P A' \qquad (25)$$

Using equations (23) and (24), the following can be obtained:

$$u_S = -R_2^{-1} \left(B^T P + R_{12}^T \right) x = -Fx \qquad (26)$$

Matrix F is then divided such that:

$$u_S = -\left[F_1 \; F_2 \; F_3 \right] \left[x_e^T \; x_d^T \; x_n^T \right] \qquad (27)$$

Now, using the integrated state space model in (3-48), the following relation is derived:

$$\left[\ddot{x}_e^T \; \dot{x}_d^T \right] = \left[A_v \; 0^2 \quad -B_v \; 0 \; A_d \; 0^2 \right] \left[x_e^T \; x \right. \qquad (28)$$

Substituting (27) into (28):

$$\left[\underline{x}_e^T \; \underline{\dot{x}}_d^T\right] = \left[\underline{A}_v \; -\underline{B}_v\underline{F}_1 \; -\underline{B}_v\underline{F}_2 \; -\underline{B}_d\underline{F}\right. \tag{29}$$

The final transfer function matrix W(s), containing the washout filters is then obtained as follows:

$$\underline{W}(s) = \left[\underline{F}_1 \; \underline{F}_2\right]\left[s\underline{I} - \underline{A}_v + \underline{B}_v\underline{F}_1 \; \underline{B}_v\underline{F}_2 \; \underline{B}_d\underline{F}_1 \; s\underline{I} - \underline{A}\right. \tag{30}$$

3. 1 Vestibular Model Implementation

First the control inputs are determined for both the simulated aircraft and the simulator platform as:

$$\underline{u}_a = \left[\beta_a^T \; \underline{a}_{Al}^T\right]^T, \underline{u}_S = \left[\beta_S^T \; \underline{a}_{SI}^T\right]^T \tag{31}$$

The outputs, which are the sensed specific force and rotation from the aircraft and the simulator platform at the pilot's head are:

$$\underline{y}_a = \left[\hat{\omega}_{Pa} \; \hat{f}_{Pa}\right], \underline{y}_S = \left[\hat{\omega}_{PS} \; \hat{f}_{PS}\right] \tag{32}$$

The transfer function for the semicircular canal can be rewritten as:

$$\frac{\hat{\omega}}{\beta} = \frac{T_L T_a s^3}{(T_a s+1)(T_L s+1)(T_s s+1)} \tag{33}$$

Where $T_L, T_a, and\, T_s$ are the semicircular canal parameters given in table (1) below:

Table 1 Semicircular Canal Parameters

	Roll(x)	Pitch(y)	Yaw(z)
$T_L[s]$	6.1	5.3	10.2
$T_s[s]$	0.1	0.1	0.1
$T_a[s]$	30	30	30

Similarly, the otolith organs can be written as a transfer function:

$$\frac{\hat{f}}{f} = \frac{K(\tau_a s+1)}{(\tau_L s+1)(\tau_s s+1)} \tag{34}$$

Where τ_a, τ_L, and τ_S are given in table (2) below:

Table 2 Otolith Organs Parameters

	Surge (x)	Sway(y)	Heave (z)
$\tau_L[s]$	5.33	5.33	5.33
$\tau_S[s]$	0.66	0.66	0.66
$\tau_a[s]$	13.2	13.2	13.2

For convenience the semicircular canals and otolith organ equations are converted into their state-space models given by:

$$\underline{\dot{x}} = \underline{A}_{scc}\underline{x} + \underline{B}_{scc}\underline{u} \;\; \underline{y} = \underline{C}_{scc}\underline{x} + \underline{D}_{scc}\underline{u} \tag{35}$$

For the otolith model, the state-space model is represented as:

$$\underline{\dot{x}} = \underline{A}_{\text{OTO}}\underline{x} + \underline{B}_{\text{OTO}}\underline{u} \;\; \underline{y} = \underline{C}_{\text{OTO}}\underline{x} + \underline{D}_{\text{OTO}}\underline{u} \tag{36}$$

Now the complete vestibular system model can be written in state-space form as:

$$\underline{A}_v = \begin{bmatrix} \underline{A}_{SCC} & \underline{0}^0 \\ \underline{0} & \underline{A}_{OTO} \end{bmatrix}, \underline{B}_v = \begin{bmatrix} \underline{B}_{SCC} & \underline{B}_{OTO} \end{bmatrix} \tag{37}$$

Which is referenced in equation (15). The complete integrated model is now ready to be evaluated.

The two matrices, $\underline{A}_d$ and $\underline{B}_d$, which remain undetermined, can be isolated. These matrices are defined by the internal additional variables $\underline{x}_d$. The state variables x are divided into three groups:

- rotational state variables $\{x_1 \cdots x_9\}$
- translational state variables $\{x_{10} \cdots x_{15}\}$
- and internal additional state variables $\{x_{16} \cdots x_{30}\}$

The model includes additional variables, such as the linear velocity and displacement of the simulator, as well as other terms. The table (3) below shows all the state variables.

Table 3 State Variables

Variable	Description	Pitch/Surge	Roll/Sway	Yaw	Heave
x_1	state variables (roll)		✓	-	-
x_2			✓		
x_3			✓		
x_4	state variables (pitch)	✓			
x_5		✓			
x_6		✓			
x_7	state variables (yaw)			✓	
x_8				✓	
x_9				✓	
x_{10}	state variable (surge)	✓			
x_{11}		✓			
x_{12}	state variable (sway)		✓		
x_{13}			✓		
x_{14}	state variable (heave)				✓
x_{15}					✓
x_{16}	$\iint \phi_S dt^2$				
x_{17}	$\int \phi_S dt$		✓		
x_{18}	$\iint \theta_S dt^2$				
x_{19}	$\int \theta_S dt$	✓			
x_{20}	$\iint \psi_S dt^2$			✓	
x_{21}	$\int \psi_s dt$			✓	
x_{22}	$\iiint a_{Sl}^x dt^3$	✓			
x_{23}	$\iint a_{Sl}^x dt^2$	✓			
x_{24}	$\int a_{Sl}^x dt$	✓			
x_{25}	$\iiint a_{Sl}^y dt^3$		✓		
x_{26}	$\iint a_{Sl}^y dt^2$		✓		
x_{27}	$\int a_{Sl}^y dt$		✓		
x_{28}	$\iiint a_{Sl}^z dt^3$				✓
x_{29}	$\iint a_{Sl}^z dt^2$				✓
x_{30}	$\int a_{Sl}^z dt$				✓

The washout matrix W(s) is determined in a number of modes. Pitch/Surge, Roll/Sway, Yaw, and Heave.

3.2 Pitch/Surge Mode:

In pitch/surge mode the state variables are:

$$\underline{x} = \left[x_4, x_5, x_6, x_{10}, x_{11}, x_{19}, x_{22}, x_{23}, x_{24}\right]^T$$

Which means the filter will be of the ninth order. The inputs are defined as:

$$\underline{u}_S = \left[\theta_S \; a_{SI}^x\right], \underline{u}_a = \left[\theta_A \; a_{AI}^x\right] \tag{38}$$

And the outputs:

$$\underline{y}_S = \left[\hat{q}_{PS}\, \hat{f}^{x}_{PS}\right], \underline{y}_A = \left[\hat{q}_{PA}\, \hat{f}^{x}_{PA}\right] \qquad (39)$$

Collecting the vestibular model, internal variables, inputs and outputs into a single state-space model gives:

$$\underline{x} = \left[\underline{x}_e^T\, \underline{x}_d^T\, \underline{x}_n^T\right]^T \quad \underline{y} = \left[\hat{q}_{PS} - \hat{q}_{PA}\, \hat{f}^{x}_{PS} - \hat{f}^{x}_{PA}\right. \qquad (40)$$

Where:

$$\underline{A} = \left[\underline{A}_v\, \underline{0} \;\; -\underline{B}_v\, \underline{0}\, \underline{A}_d\, \underline{0}^2\, \underline{0}\, \underline{0}\, \underline{A}_n\right], \underline{B} = \left[\underline{B}_v\, \underline{B}_{\ldots}\right. \qquad (41)$$

To minimize sensation error, the cost function given in equation (21) is used and W(s) is obtained from equation (30). The MATLAB Riccati equation solver is employed to obtain the 9th order filters.

3.3 Roll/Sway Mode:

State variables are:

$$\underline{x} = \left[x_1, x_2, x_3, x_{12}, x_{13}, x_{17}, x_{25}, x_{26}, x_{27}\right]^T$$

And the inputs are:

$$\underline{u}_S = \left[\phi_S\, a^{y}_{SI}\right], \underline{u}_a = \left[\phi_A\, a^{y}_{AI}\right] \qquad (42)$$

The outputs are represented as:

$$\underline{y}_S = \left[\hat{p}_{PS}\, \hat{f}^{y}_{PS}\right], \underline{y}_A = \left[\hat{p}_{PA}\, \hat{f}^{y}_{PA}\right] \qquad (43)$$

Note the symbols y indicating the direction of the accelerations and angular velocities. The vestibular system model will be similar to that in pitch/surge mode but will use the constants from tables (1) and (2) corresponding to the roll and sway values. Similarly, the full integrated system is acquired as in pitch/surge mode using the modified inputs, outputs, and state variables. The washout filters are then evaluated using MATLAB.

4. Results and Discussion

To begin, the algorithm was tested with white noise as an input (as in the model development) because it closely resembles motion signals from the aircraft in different situations, thus giving a good idea of how the algorithm will behave in real time testing. For pitch surge mode, the input white noise is shown in figure (3), and (4) shows the high-pass filter response in surge. The response of the high-pass filter is as expected, only picking up high frequency changes, but due to the nature of the input white noise, almost all changes are high in frequency which results in the high pass surge closely following the input only with lower magnitudes due to the scaling and limiting.

Next, the response in both Pitch/Surge, and Roll/Sway modes is tested using the vestibular system. The inputs from the simulated vehicle and the outputs from the filter channels are both used as inputs to the vestibular model for each mode. This is to extract the human sensation for each response and to gauge the error between actual sensation and platform sensation.

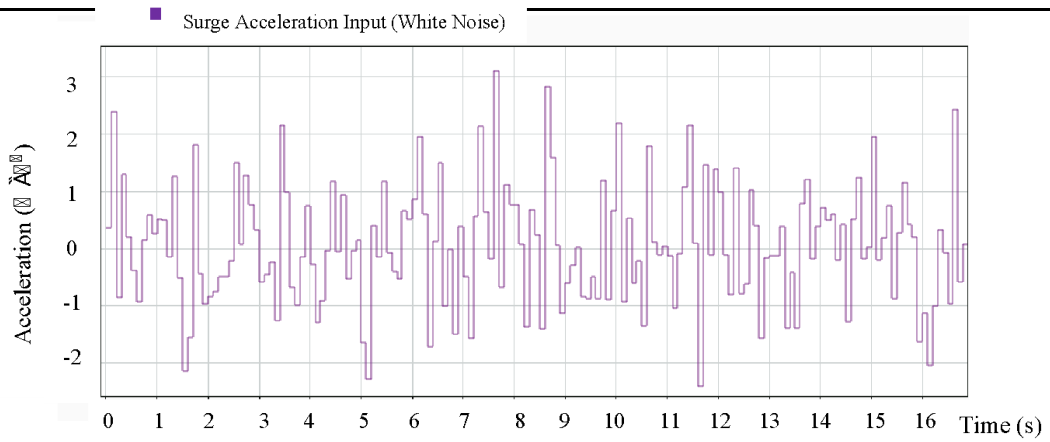

Figure 3 Input White Noise as Surge Acceleration

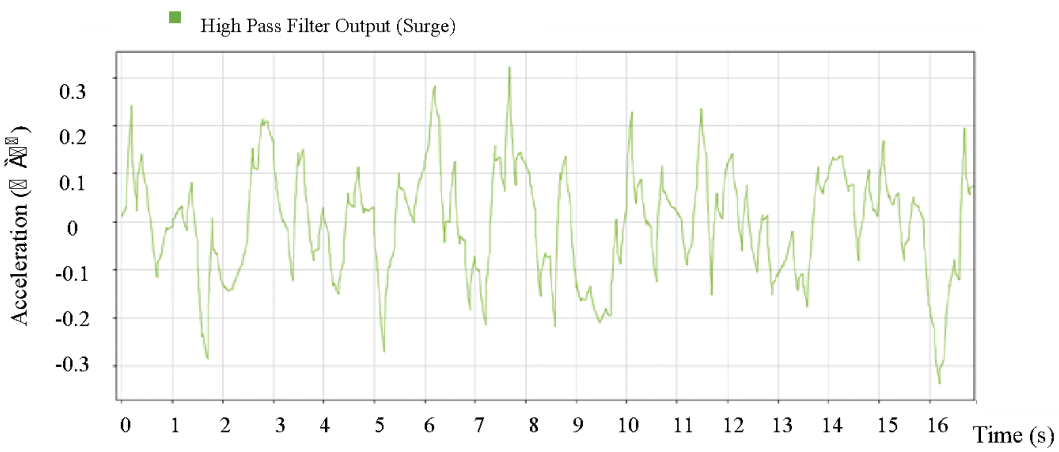

Figure 4 High Pass Filter Response in Surge

In figure (5), it can be seen that the sensation provided by the optimal washout filter in surge is pretty accurate to the real aircraft sensation. Sometimes, however, the response lags or is exaggerated by the MCA but overall, the sensation is rather reasonable. The same sort of response is seen in the Roll/Sway mode, figure (6) shows the sensed platform sway tracking the sensed aircraft sway pretty closely and accurately although sometimes lacking in magnitude. For example, at the beginning the biggest difference between the signals can be seen when the sensed platform acceleration is at 1.5 m/s^2 and the sensed vehicle acceleration is higher at about 2.8 m/s^2.

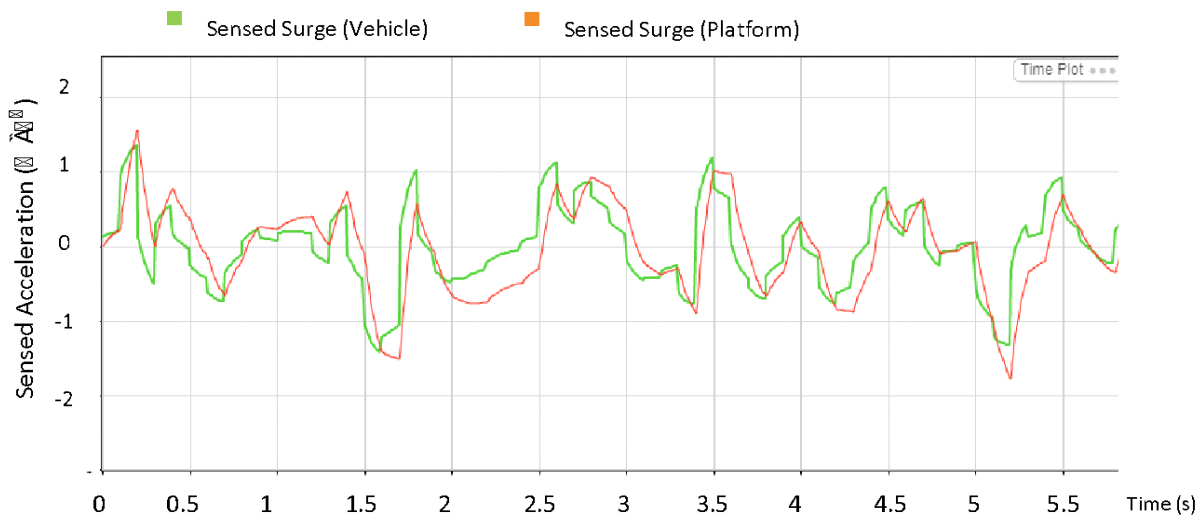

Figure 5 Vehicle vs Platform Sensed Surge (LQR)

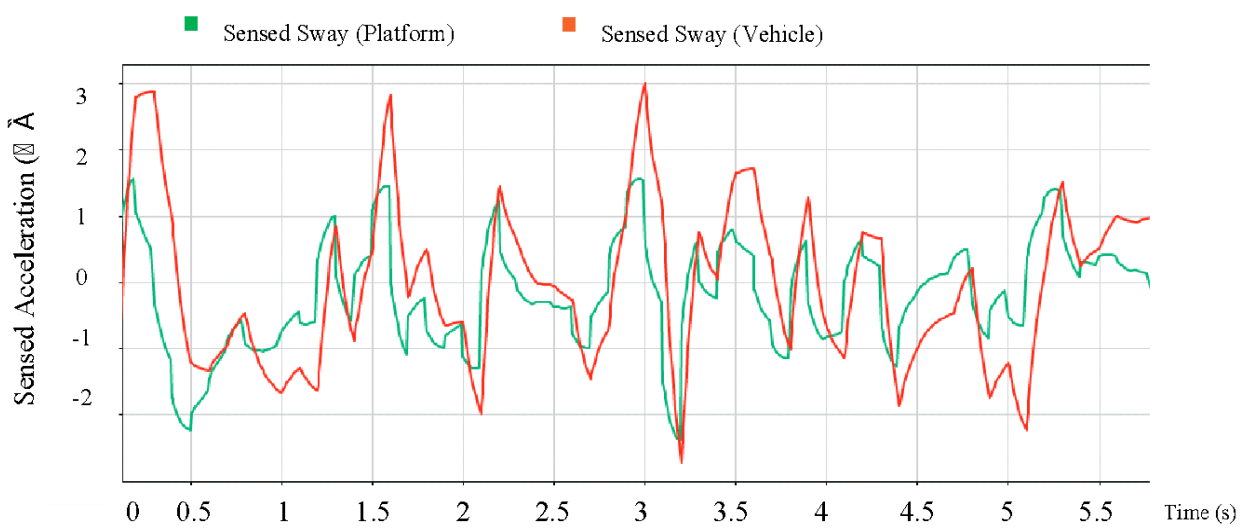

Figure 6 Vehicle vs Platform Sensed Sway (LQR)

Now to compare the response of the optimal washout filter to that of the classical algorithm of [8] , the same approach will be used. The outputs of the classical filter will be passed through the vestibular model in order to obtain the relevant sensations. This is also done in both pitch/surge and roll/sway modes. Figure (7) below shows the classical washout filter sensation output in surge when the inputs are the same white noise shown previously in figure (3). When compared with the output of the optimal filter in figure (5), it can be seen that the sensation error or difference is much more pronounced although the accelerations do track that of the real vehicle to a reasonable extent. This is a direct effect of the worst-case based tuning of the algorithm which makes it behave well in some cases but below average in most. In roll/sway mode, the sensation error remains high, but in much the same fashion, the tracking is good and with some tuning the results will be much more accurate. Figure (8) provides some context that suggests that the only tuning needed is in the scaling of the output sensation from the filter when compared to that of the optimal filter in figure (6).

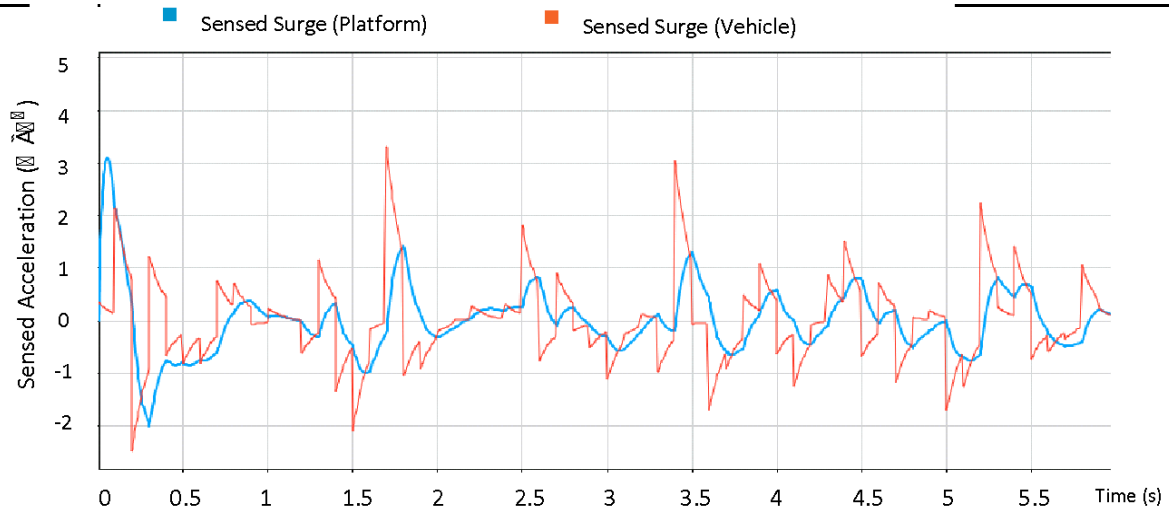

Figure 4 Vehicle vs Platform Sensed Surge (Classical)

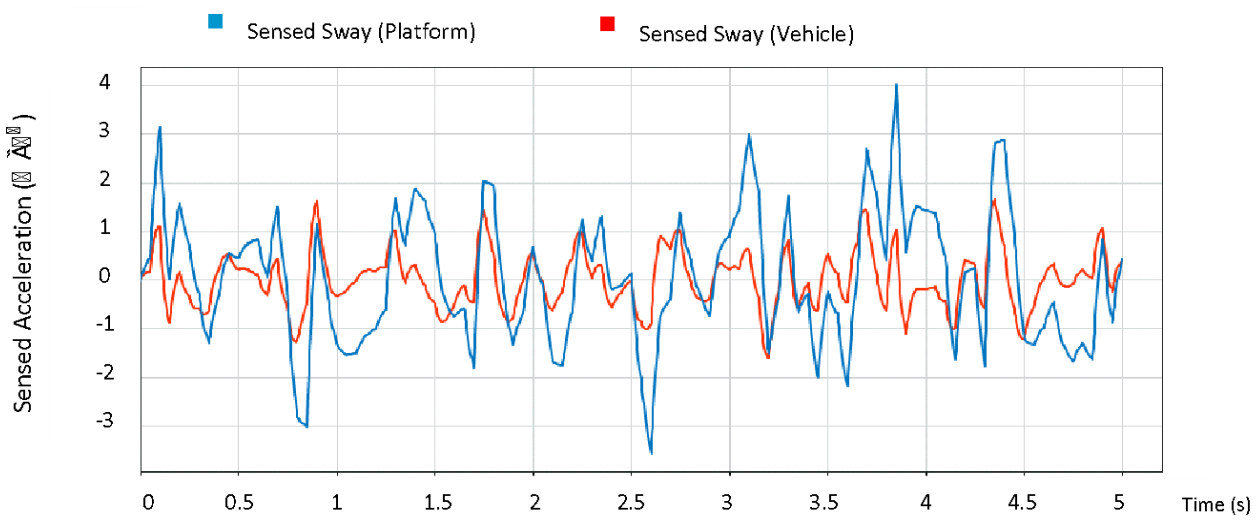

Figure 8 Vehicle vs Platform Sensed Surge (Classical)

The final test is done in real-time by utilizing the interfacing setup developed in [8] between the simulation environment and Simulink where the algorithms are implemented. For this test, a new take-off scenario is used and the results are shown in figures (9) and (10). Once again, the classical filter is compromised in its fidelity due to the nature of the parameter tuning method and provides very little in terms of sensation accuracy, and responds to high frequency changes in acceleration only. The optimal filter, on the other hand is rather accurate although conservative in intensity. Both of the filters however display the effect of the washout in that they always attempt to return to the neutral position.

Referring to figures (11) and (12), the maximum sensation error in surge for the optimal and classical algorithms can be seen. It is clear that the sensation error, although seemingly large in both cases, shows that the optimal washout filter is much more accurate to the actual aircraft surge, showing a maximum difference of 208.7 m/s^2 and an difference of about 85 m/s^2. For the classical filter, on the other hand, it had a maximum difference of 360.35 m/s^2, and an average difference of about 165 m/s^2.

5. Conclusion

The present study presented the outcomes of the optimal washout filter MCA and offered a comprehensive evaluation of pilot sensation by comparing it with the classical approach. The results demonstrate a distinct advantage of the optimal washout filter approach, emphasizing the impact of formulating the MCA as an optimal design problem and incorporating human sensation into the design process. The utilized control algorithm was founded on the Linear Quadratic Regulator (LQR) and entailed a state feedback controller that integrated the dynamics of the platform, limitations, and a model of human vestibular sensation. The objective of this was to develop motion cues that are both reliable and

precise, thus enabling the creation of a more robust simulator platform.

The algorithm was executed and evaluated, with the outcomes revealing significant improvements in the sensation experienced by the individual compared to the traditional motion cueing approach. This was demonstrated by a marked and thorough reduction in the average sensation

error, with a value of 85 m/s^2 in contrast to the previously observed 165 m/s^2.

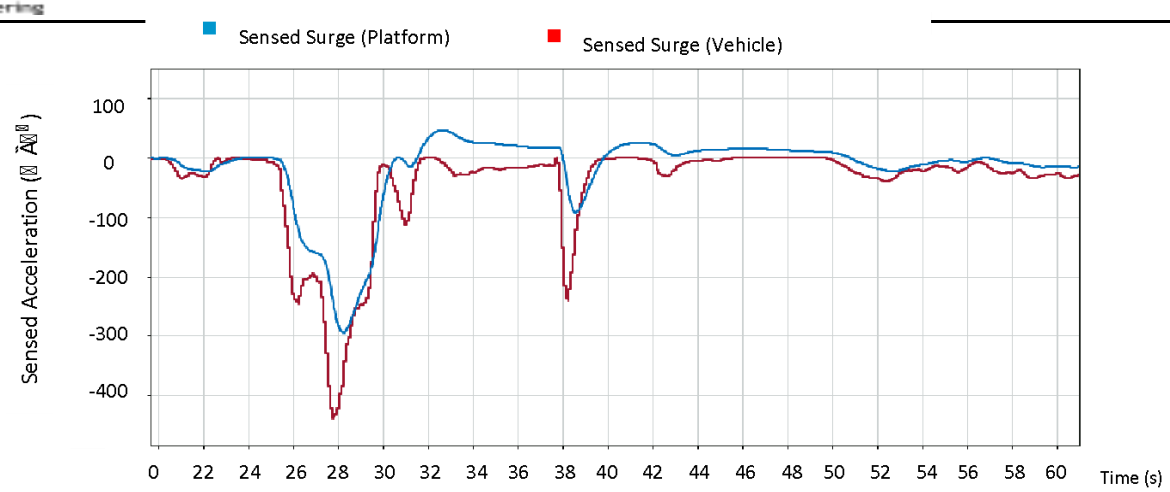

Figure 9 Optimal Washout Filter Sensation in Take-off Scenario

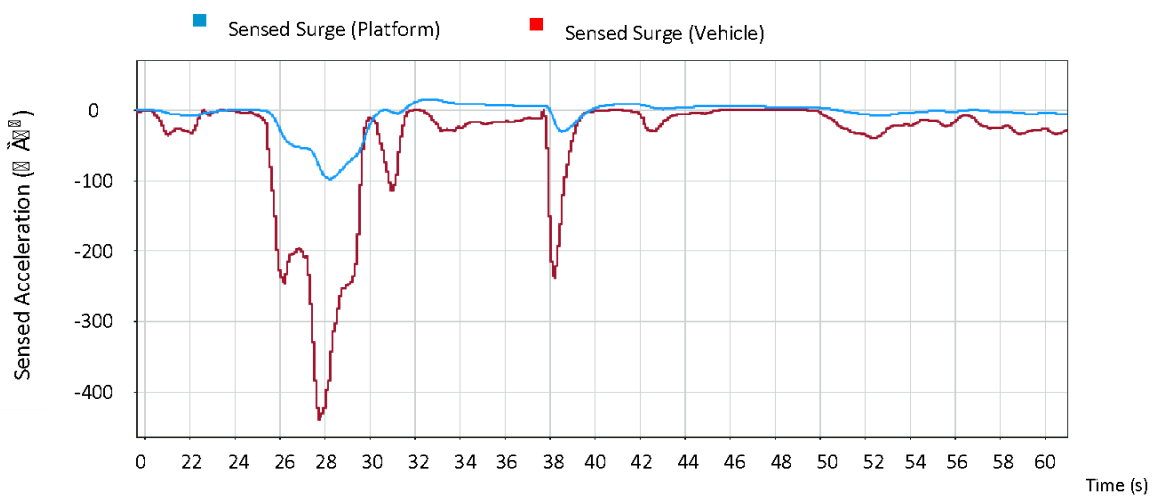

Figure 10 Classical Washout Filter Sensation in Take-off Scenario

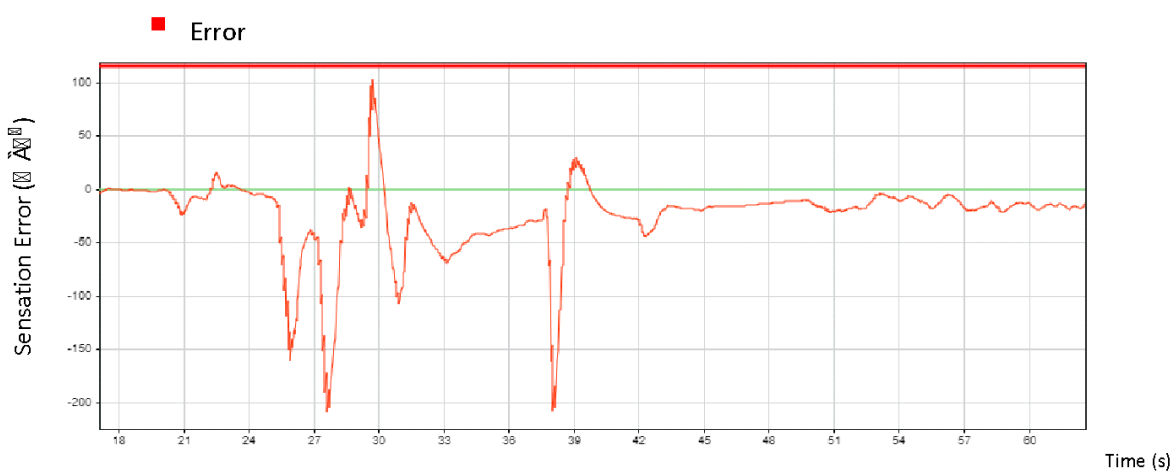

Figure 11 Sensation Error in Surge (Optimal)

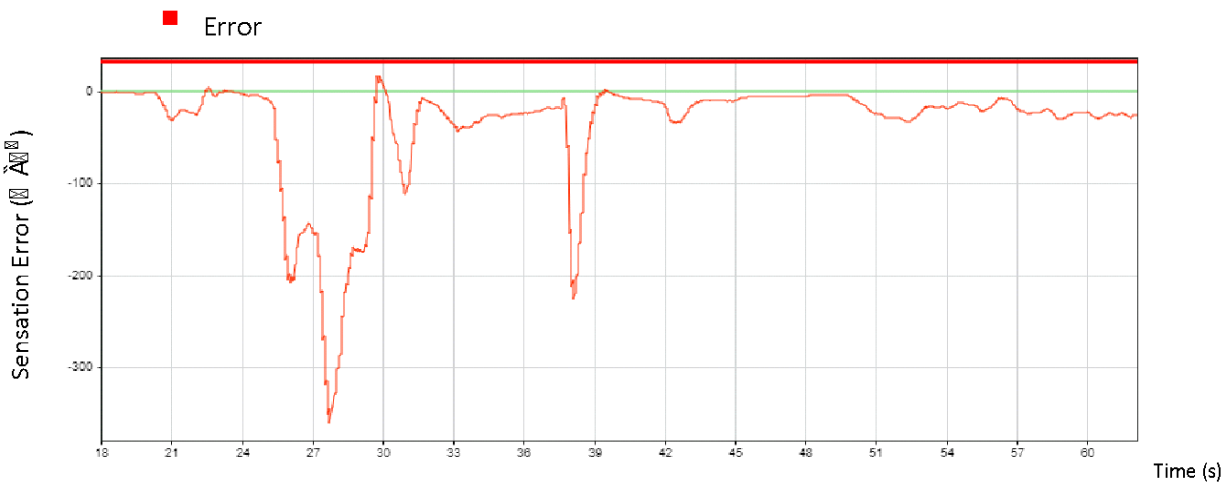

Figure 12 Sensation Error in Surge (Classical)

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Declarations

Conflict of interest The authors report no conflicts of interest. The authors alone are responsible for the content and writing of this research.

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